

Mathematical Foundations for Machine Learning - Mid-Semester 1 Exam Guide

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MFML Midsem 1 Preparation Hub

Course: Mathematical Foundations for Machine Learning. Exam: 19 September 2026, closed book, 90 minutes, 30 marks. Scope: Contact Sessions 1-8.

Schedule source: the course-local [2026 academic calendar record](#) gives AMLCS ZC416, 19 September 2026, 09:00 AM-11:00 AM in the Regular Examination Schedule, EC2 table. It mirrors the later official schedule supplied and recorded by the repository owner. The course handout supplies the 90-minute, closed-book format and Sessions 1-8 scope, but its examination date is "To be announced"; it does not support the calendar date.

Begin with the [syllabus map](#), then use the session cards for fast learning and this guide for practice. The guide and its manifested source files assume only high-school algebra and are standalone. A printable copy is the [exam guide PDF](#).

Six-day plan

Day 1 - 13 September: systems and spaces

Learn [Sessions 1-2](#), then read the exact [Session 1 card](#) and [Session 2 card](#). Practise [C01-C06](#) and [N01-N06](#). Check only after completing a page of working.

Day 2 - 14 September: analytic geometry

Learn [Session 3](#), then read the exact [Session 3 card](#). Practise [C07-C09](#) and [N07-N09](#). Recalculate every dot product and norm.

Day 3 - 15 September: decompositions

Learn [Sessions 4-5](#), then read the exact [Session 4 card](#) and [Session 5 card](#). Practise [C10-C15](#) and [N10-N15](#).

Day 4 - 16 September: derivatives and graphs

Learn [Sessions 6-7](#), then read the exact [Session 6 card](#) and [Session 7 card](#). Practise [C16-C19](#) and [N16-N19](#).

Day 5 - 17 September: curvature and integration

Learn [Session 8](#), then read the exact [Session 8 card](#). Practise [C20-C24](#) and [N20-N24](#). Attempt the [historical-source-aware items](#) without treating third-party metadata as official.

Day 6 - 18 September: timed rehearsal and recall

Sit [Mock 01](#), or its [printable PDF](#), for 90 minutes. Only after submitting your attempt, open the [answer key](#), [point rubric](#), or [solutions PDF](#). Record lost marks by session, redo only those topics, and finish with the [last-30-minutes route](#).

How to write a calculation answer

Show the governing formula, substitute the supplied values, include non-obvious steps, state shapes and conditions, and interpret the result. A numerical answer without its conditions can be mathematically incomplete.

MFML Midsem 1 Syllabus Map and Mini Lessons

The handout prescribes Sessions 1-8. "Local evidence" means a matching lecture artifact exists in this repository; it does not change prescribed scope.

Session	Prescribed topic	Local evidence
1	Linear systems and matrices	Strong
2	Vector spaces, independence, span, basis, rank/nullity, affine spaces	Strong except affine spaces
3	Norms, inner products, distance, angles, orthogonality, orthonormal bases	Not substantive
4	Determinant, trace, eigenpairs, Cholesky	Determinant fragment only
5	Diagonalization, SVD, matrix approximation	Not substantive
6	Univariate and partial derivatives, gradients, Jacobians	Not substantive
7	Matrix gradients, backpropagation, automatic differentiation	Not substantive
8	Hessians, linearization, multivariate Taylor series, unconstrained extrema and review	Not substantive

Prerequisite bridge

A scalar is one number. A vector is an ordered list of numbers, normally written as a column. A matrix is a rectangular array. "Shape m by n " means m rows and n columns. Matrix multiplication is allowed only when the touching dimensions match: an m by n matrix times an n by p matrix produces an m by p matrix.

Superscript T means transpose, which swaps rows and columns. $\mathbb{R}^n$ means all real columns with n entries. $\sqrt{a}$ is the nonnegative number whose square is a . A function maps an input to an output; a derivative describes how that output changes after a small input change.

Sessions 1-2: systems and spaces

Session 1 - Linear systems and matrices

A linear equation such as $2x+y=5$ contains unknowns only to the first power. Several equations can be packed into $Ax=b$. If A is m by n , then the unknown x is n by 1 and the result b is m by 1 .

Row operations preserve the solution set: swap rows, multiply a row by a nonzero scalar, or add a multiple of one row to another. A pivot is a leading nonzero entry after reduction. Rank counts independent pivot directions. The augmented matrix $[A|b]$ appends b to A .

$\text{rank}(A) < \text{rank}([A b])$	no solution
$\text{rank}(A) = \text{rank}([A b]) = n$	one solution
$\text{rank}(A) = \text{rank}([A b]) < n$	infinitely many solutions

An inverse can solve $Ax=b$ as $x=A^{-1}b$ only when A is square and nonsingular. For a square matrix, nonsingular, invertible, full rank, and nonzero determinant are equivalent conditions.

Worked illustration. In $x+y=2$, $2x+2y=4$, the second augmented row is twice the first. Both ranks are 1, so the system is consistent. Since $n=2$, one variable is free. Setting $y=t$ gives $x=2-t$, and substitution checks both equations.

Session 2 - Vector spaces and affine spaces

A linear combination $c_1v_1+\dots+c_kv_k$ is a weighted sum of vectors. The span is the set of every such sum. Vectors are linearly independent when the zero vector can be produced only with all coefficients zero. A basis both spans the space and is independent.

A subspace contains zero and is closed under addition and scalar multiplication. An affine space $a+U$ shifts a subspace U by a fixed vector a ; it need not contain zero. It is a subspace exactly when a belongs to U . An affine combination has coefficients summing to one. A convex combination additionally requires every coefficient to be nonnegative.

For an m by n matrix, the column space lies in $\mathbb{R}^m$, the row space lies in $\mathbb{R}^n$, the null space contains solutions of $Ax=0$, and the left-null space contains solutions of $A^T y=0$. If $\text{rank}(A)=r$, all four dimensions are:

A is m by n and $\text{rank}(A) = r$

$$\dim \text{Col}(A) = r$$

$$\dim \text{Row}(A) = r$$

$$\dim \text{Null}(A) = n - r$$

$$\dim \text{LeftNull}(A) = m - r$$

input directions: $r + (n-r) = n$; output directions: $r + (m-r) = m$

The nullity is $n-r$, the number of free input directions solving $Ax=0$. The left nullity is $m-r$, the number of output-side directions solving $A^T y=0$. Rank-nullity says

$$\text{rank}(A) + \text{nullity}(A) = n$$

To build a column-space basis, row-reduce to identify pivot positions but take the corresponding columns from the original matrix; row operations generally change the column space.

Worked illustration. The columns $v_1=(1,0)$ and $v_2=(1,1)$ are independent: $c_1v_1+c_2v_2=0$ forces $c_2=0$, then $c_1=0$. Any (a,b) equals $(a-b)v_1+bv_2$, so the pair also spans $\mathbb{R}^2$ and is a basis. By contrast, $(1,0)+\text{span}((0,1))$ is the affine line $x=1$ and excludes zero.

Session 3: analytic geometry

A norm measures length. For $x=(x_1, \dots, x_n)$, the 1-norm adds absolute values, the 2-norm is ordinary Euclidean length, and the infinity-norm is the largest absolute entry.

$$\|x\|_1 = |x_1| + \dots + |x_n|$$

$$\|x\|_2 = \sqrt{x_1^2 + \dots + x_n^2}$$

$$\|x\|_\infty = \max_i |x_i|$$

$$d(x, y) = \|x - y\|_2$$

The standard inner product is $x^T y$, the sum of coordinate products. It is symmetric, linear, and positive definite: $x^T x \geq 0$, with equality only for zero. Cauchy-Schwarz states

$$|x^T y| \leq \|x\|_2 \|y\|_2$$

That bound keeps the following ratio between -1 and 1, so it can be a cosine. Both vectors must be nonzero.

$$\cos(\theta) = \frac{x^T y}{\|x\|_2 \|y\|_2}$$

Orthogonal vectors have dot product zero. Orthonormal vectors are orthogonal and each has norm 1. For nonzero u , suppose the closest vector along its line is cu . Orthogonality of residual $x - cu$ to u requires $u^T(x - cu) = 0$, so $c = (u^T x) / (u^T u)$. This derives the projection formula.

$$\text{proj}_u(x) = u \frac{u^T x}{u^T u}$$

If the columns of Q are orthonormal, $Q^T Q = I$ and the projection onto their span is $Q Q^T x$. The formula $Q^{-1} = Q^T$ additionally requires a real square orthogonal Q .

Gram-Schmidt builds orthonormal columns one input vector at a time. Suppose $q_1, \dots, q_{j-1}$ are already orthonormal. For the next input v_j , each coefficient $q_i^T v_j$ measures how much of v_j points along q_i . Subtract all those projections to obtain the residual r_j , then normalize it:

$$r_j = v_j - \sum_{i=1}^{j-1} q_i (q_i^T v_j)$$

$$q_j = r_j / \|r_j\|_2, \text{ provided } r_j \neq 0$$

$$r_j = 0 \Leftrightarrow v_j \text{ is already in span}(q_1, \dots, q_{j-1})$$

The subtraction removes every earlier direction, so r_j is orthogonal to $q_1, \dots, q_{j-1}$. If r_j is nonzero, $q_j = r_j / \|r_j\|_2$ has length one and adds an independent direction. If $r_j = 0$, then v_j was already in the earlier span; normalization is impossible, and the original input list is dependent.

In floating-point computation, modified Gram-Schmidt or a tested QR routine is usually more numerically stable than a literal classical Gram-Schmidt implementation.

Worked illustration. For $x=(3,2)$ and $u=(1,1)$, $u^T x=5$ and $u^T u=2$. The projection is $(5/2, 5/2)$. The residual $(1/2, -1/2)$ has dot product zero with u .

Sessions 4-5: matrix decompositions

Session 4 - Determinant, trace, eigenpairs, and Cholesky

For the matrix below, the determinant is $ad-bc$. A square matrix has an inverse exactly when its determinant is nonzero. Trace is the sum of diagonal entries.

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

An eigenvector is a nonzero direction v that a square matrix scales without turning: $Av = \lambda v$. The scale λ is its eigenvalue. An eigenbasis is a basis made entirely of eigenvectors. A real symmetric matrix, meaning $A^T = A$, has real eigenvalues and an orthonormal eigenbasis.

A real symmetric matrix is positive definite when $x^T A x > 0$ for every nonzero real x . Ordinary Cholesky applies exactly to such matrices and writes $A = LL^T$, where L is lower triangular with positive diagonal. If A is m by n and B is n by m , then both square products exist and $\text{tr}(AB) = \text{tr}(BA)$: the two traces sum the same products $a_{ij} b_{ji}$ in a different order.

$$A = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 1 & \sqrt{2} \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & \sqrt{2} \end{bmatrix} = LL^T$$

Multiplying the factors reconstructs every entry of A . Symmetry alone is insufficient: $\text{diag}(1, -1)$ is symmetric but not positive definite.

Eigenpair illustration. The following direct products give eigenvalues 3 and 1; the trace 4 equals their sum and the determinant 3 equals their product.

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

trace = 4 = 3+1, determinant = 3 = 3·1

Session 5 - Diagonalization, SVD, and low-rank approximation

Diagonalization writes a square matrix as $A = P D P^{-1}$, where columns of P form a full eigenbasis and D stores matching eigenvalues. It exists exactly when an n by n matrix has n independent eigenvectors. Distinct eigenvalues are sufficient, not necessary. A zero eigenvalue can occur in a diagonalizable matrix and makes it singular.

When that full eigenbasis exists, P is invertible and powers are easy because adjacent $P^{-1} P$ factors cancel:

$$A = P D P^{-1}, \text{ with } P \text{ invertible}$$

$$A^2 = P D (P^{-1} P) D P^{-1} = P D^2 P^{-1}$$

$$\text{Therefore } A^k = P D^k P^{-1} \text{ for } k = 0, 1, 2, \dots$$

The identity applies to nonnegative integer powers k ; a negative power additionally requires A , hence D , to be invertible.

Every real m by n matrix has a singular value decomposition. Its columns are orthonormal directions and its singular values are nonnegative scale factors.

$$A = U \Sigma V^T$$

Positive singular values count the rank. The spectral norm $\|A\|_2$ is the largest singular value. The Frobenius norm $\|A\|_F$ is the square root of the sum of all squared singular values.

Order the positive singular values $\sigma_1 \geq \dots \geq \sigma_r > 0$. Let A_k retain the first k singular directions. Eckart-Young says A_k is a best rank-at-most- k approximation and gives both errors:

$$\|A - A_k\|_2 = \sigma_{k+1}$$

$$\|A - A_k\|_F = \sqrt{(\sigma_{k+1}^2 + \dots + \sigma_r^2)}$$

Here σ_{k+1} is the first omitted singular value and $r = \text{rank}(A)$. If $k \geq r$, both errors are zero. Nonuniqueness can occur at a positive truncation tie: when $k < r$ and $\sigma_k = \sigma_{k+1} > 0$, different choices inside the tied singular subspace attain the same minimum error. A tie only among omitted zeros does not create a distinct zero-error approximant.

Right singular vectors are eigenvectors of $A^T A$; for positive singular values, the matching left singular vectors are eigenvectors of $A A^T$.

Worked illustration. If singular values are 7, 3, 2, the best rank-1 spectral error is 3, while its Frobenius error is $\sqrt{3^2 + 2^2} = \sqrt{13}$. For rank 2, both errors are 2.

Sessions 6-7: derivatives and computation graphs

Session 6 - Derivatives, gradients, and Jacobians

For one variable, $f'(x)$ is the limiting slope and estimates $f(x+h) - f(x)$ by $f'(x)h$ for small h . A partial derivative changes one input while holding the others fixed. For scalar $f: \mathbb{R}^n \rightarrow \mathbb{R}$, the gradient is the n by 1 column of partial derivatives.

Three basic rules are: the derivative of e^x is e^x , the derivative of $\sin(x)$ is $\cos(x)$, and the derivative of $\cos(x)$ is $-\sin(x)$. For products, differentiate one factor at a time: $(uv)' = u'v + uv'$. In a partial derivative, treat the other variables as constants. Repeating the process gives second partial derivatives for a Hessian and Taylor model.

$$d(e^x)/dx = e^x; d(\sin x)/dx = \cos x; d(\cos x)/dx = -\sin x$$

$$h(x,y) = e^x \sin y$$

$$\text{partial } h / \text{partial } x = e^x \sin y; \text{partial } h / \text{partial } y = e^x \cos y$$

$$\text{at } (0,0): h = 0, \text{grad } h = (0,1), \text{Hessian rows} = (0,1), (1,0)$$

$$\text{second-order Taylor model near zero: } h(x,y) \approx y + xy$$

For the illustration $h(x,y)=e^x \sin(y)$, at $(0,0)$ the value is 0, the gradient is $(0,1)$, and the Hessian has rows $(0,1)$ and $(1,0)$. Substitution into the second-order Taylor formula gives $h(x,y)$ approximately $y+xy$ near zero. This is a local polynomial model, not an identity for distant points.

For a unit direction u , the directional derivative is $u^T \text{grad } f$. Cauchy-Schwarz bounds it by $\|\text{grad } f\|_2$; when the gradient is nonzero, equality is achieved by the unit direction $u = \text{grad } f / \|\text{grad } f\|_2$. Thus the gradient gives steepest local increase under Euclidean length. This claim is local and assumes differentiability.

For vector output $g: \mathbb{R}^n \rightarrow \mathbb{R}^m$, this guide uses the output-by-input Jacobian convention: row i , column j is $\text{partial } g_i / \text{partial } x_j$, so J_g is m by n .

$$g(x+h) \approx g(x) + Jg(x)h$$

For $g(x,y)=(x^2+y,xy)$,

$$Jg(x,y) = \begin{bmatrix} 2x & 1 \\ y & x \end{bmatrix}$$

At $(1,2)$, the Jacobian and a small displacement calculation are:

$$Jg(1,2)h = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 0.02 \\ -0.01 \end{bmatrix} = \begin{bmatrix} 0.03 \\ 0.03 \end{bmatrix}$$

Session 7 - Matrix gradients, backpropagation, and autodiff

Using column vectors and scalar outputs, useful identities are

$$\text{grad}_x(a^T x) = a$$

$$\text{grad}_x(x^T P x) = (P+P^T)x$$

$$\text{grad}_x((1/2)\|Ax-b\|_2^2) = A^T(Ax-b)$$

The shortcut $2Px$ needs symmetric P . The least-squares identity follows by setting residual $r=Ax-b$; a small change h gives r change Ah , so the loss change is approximately $r^T Ah = (A^T r)^T h$.

A computation graph stores elementary operations and intermediate values. Backpropagation is reverse-mode chain rule on that graph. Automatic differentiation is neither symbolic algebra nor finite

differences: it evaluates exact local derivative rules for operations actually executed.

For vector output y with Jacobian J_y , reverse mode needs a seed column v matching the output shape. It returns the vector-Jacobian product shown below, which combines output sensitivities into input sensitivities.

$$\text{input sensitivity} = J_y^T v$$

Worked illustration. For $u=x^2$, $y=2u+1$, $L=y^2$ at $x=1$, the forward values are $u=1$, $y=3$, $L=9$. Starting with scalar seed 1, reverse multiplication gives $6*2*2=24$.

Session 8: Hessians, Taylor models, and extrema

The Hessian of scalar $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is the n by n matrix of second partial derivatives. Linearization uses the gradient only; the quadratic Taylor model also uses the Hessian.

$$\text{first order: } f(x+h) \approx f(x) + \text{grad } f(x)^T h$$

$$\text{second order: } f(x+h) \approx f(x) + \text{grad } f(x)^T h + (1/2) h^T H(x) h$$

These are local approximations for sufficiently small h under suitable differentiability. At a differentiable interior unconstrained local extremum, zero gradient is necessary, not sufficient. At a stationary point:

- positive-definite Hessian gives a strict local minimum;
- negative-definite Hessian gives a strict local maximum;
- indefinite Hessian gives a saddle;
- positive- or negative-semidefinite Hessian can be inconclusive.

For example, x^6 and $-x^6$ both have zero first and second derivatives at zero but have opposite local behavior. A local minimum is global only with additional whole-domain reasoning. A convex function lies below every straight chord joining two points on its graph; consequently any local minimum is global. Strict convexity gives at most one minimizer.

Worked illustration. For $f(x,y)=x^2+3y^2-2x+6y$, the complete stationary-point calculation is:

$$\begin{aligned} \text{grad } f(x,y) &= \begin{bmatrix} 2x-2 \\ 6y+6 \end{bmatrix} = 0 \Rightarrow (x,y)=(1,-1) \\ H &= \text{diag}(2,6), \quad z^T H z = 2z_1^2 + 6z_2^2 > 0 \text{ for } z \neq 0 \\ f(x,y) &= (x-1)^2 + 3(y+1)^2 - 4 \end{aligned}$$

The Hessian is positive definite, so the point is a strict local minimum; the completed-square form proves the stronger unique global minimum and value -4.

Restored mixed practice with explained answers

Use these as illustrations, then attempt the answer-free banks.

- 1 For $x=(1,2,2)$, $\|x\|_2 = \sqrt{1+4+4}=3$.

- 2 Projecting $(2,1)$ onto $(1,0)$ gives $(2,0)$ and perpendicular residual $(0,1)$.
- 3 The plane $x+y+z=1$ is affine but not a linear subspace because it excludes zero.
- 4 Orthonormal columns satisfy $Q^ATQ=I$, so $\|Qx\|^2=x^ATQ^ATQx=\|x\|^2$: multiplication by Q preserves Euclidean length.
- 5 The eigenvalues of $\text{diag}(5,-1,2)$ are its diagonal entries: $5,-1,2$.
- 6 Singular values cannot be negative because they are defined as nonnegative square roots of eigenvalues of A^TA .
- 7 For $(1/2)\|x-b\|^2$, the gradient is $x-b$; this is the least-squares identity with $A=I$.
- 8 The origin of $f(x,y)=x^2-4y^2$ is a saddle because the Hessian has one positive and one negative eigenvalue.
- 9 Reconstruction checks are $LL^T=A$ for Cholesky and $U\Sigma V^T=A$ plus orthonormality for SVD.

Integrated exam checklist

Before accepting a result, ask: Are shapes compatible? Is the governing formula shown? Are all required conditions stated? Are substitutions and intermediate steps visible? Is a conclusion merely local or actually global? Does the decomposition exist for this matrix class? Has the result been interpreted?

Conceptual Drills

These prompts are answer-free. Use the mini lessons in the [syllabus map](#) before attempting them.

- 1 C01 - Session 1. State a complete rank-based decision rule for zero, one, or infinitely many solutions of $Ax=b$.
- 2 C02 - Session 1. State all three elementary row operations and why a zero row-scaling factor is forbidden.
- 3 C03 - Session 1. Distinguish $Ax=b$ from $AX=B$ using shapes.
- 4 C04 - Session 2. Define span, linear independence, and basis, and state how the three ideas connect.
- 5 C05 - Session 2. Describe how row reduction is used to select a basis for the column space, specifying which matrix supplies the final vectors.
- 6 C06 - Session 2. Compare a subspace U with an affine set $a+U$; include the role of zero.
- 7 C07 - Session 3. Distinguish orthogonal from orthonormal vectors.
- 8 C08 - Session 3. State all conditions needed for the angle formula and for projection QQ^Tx .
- 9 C09 - Session 3. During Gram-Schmidt a new residual is zero. What conclusion, if any, follows about the input list?
- 10 C10 - Session 4. Relate determinant, rank, a zero eigenvalue, and invertibility for a square matrix.
- 11 C11 - Session 4. Define an eigenpair and explain why the eigenvector must be nonzero.
- 12 C12 - Session 4. State the exact matrix class for ordinary real Cholesky and contrast symmetry with positive definiteness.

- 13 C13 - Session 5. Give the necessary-and-sufficient condition for diagonalizability and one sufficient shortcut.
- 14 C14 - Session 5. Contrast eigenvalues and singular values, including signs and matrix shapes.
- 15 C15 - Session 5. State the Eckart-Young conclusion in spectral and Frobenius norms, including what a tie can change.
- 16 C16 - Session 6. Declare the Jacobian convention for $g: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and compare it with the gradient of a scalar.
- 17 C17 - Session 6. State what a first-order linearization estimates and the regime in which its error claim applies.
- 18 C18 - Session 7. Contrast automatic differentiation with symbolic differentiation and finite differences.
- 19 C19 - Session 7. Explain why a non-scalar reverse-mode calculation needs a seed vector.
- 20 C20 - Session 8. State the first-order necessary condition for an unconstrained local extremum, with every qualifier.
- 21 C21 - Session 8. Give all Hessian definiteness classifications at a stationary point.
- 22 C22 - Session 8. What conclusions are justified by a semidefinite Hessian at a stationary point? Support the answer with examples.
- 23 C23 - Integrated. Give an example of a matrix that is diagonalizable but singular and explain both properties.
- 24 C24 - Integrated. Build a five-item condition checklist that prevents common MFML exam errors.

Numerical Drills

These prompts are answer-free. For each, show formula, substitution, intermediate steps, shapes or domains, conditions, and interpretation.

- 1 N01 - Session 1. Solve $x+y=7$, $2x-y=2$, then substitute into both equations.
- 2 N02 - Session 1. Classify $x+y=2$, $2x+2y=5$ using both ranks.
- 3 N03 - Session 1. Classify $x+2y=4$, $2x+4y=8$ and parameterize every solution.
- 4 N04 - Session 2. Test whether $(1,0,1)$, $(0,1,1)$, $(1,1,2)$ are independent.
- 5 N05 - Session 2. A 4 by 7 matrix has rank 3 . Find nullity and the dimensions of all four fundamental subspaces.
- 6 N06 - Session 2. Decide whether the line $(2,1)+t(1,-1)$ is a subspace of $\mathbb{R}^2$.
- 7 N07 - Session 3. For $x=(3,-4)$, compute the 1 -, 2 -, and infinity-norms.
- 8 N08 - Session 3. Find the angle between $(1,1)$ and $(1,-1)$ and state why the formula is defined.
- 9 N09 - Session 3. Project $(3,5)$ onto the span of unit vector $(1,0)$ and verify residual orthogonality.
- 10 N10 - Session 4. For the matrix below, compute determinant, trace, and eigenvalues.

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

- 1 N11 - Session 4. Compute the Cholesky factor of the matrix below, then reconstruct the matrix.

$$\begin{bmatrix} 9 & 3 \\ 3 & 5 \end{bmatrix}$$

- 1 N12 - Session 4. Determine whether $\text{diag}(0,5)$ is invertible and diagonalizable; justify both separately.
- 2 N13 - Session 5. Use diagonalization to compute the fifth power of $\text{diag}(2,-1)$.
- 3 N14 - Session 5. For singular values $6,2,1$, calculate the best rank-1 and rank-2 spectral errors.
- 4 N15 - Session 5. For singular values $6,2,1$, calculate the best rank-1 and rank-2 Frobenius errors.
- 5 N16 - Session 6. Compute the gradient of $f(x,y)=x^2+3xy+y^2$ at $(1,2)$.
- 6 N17 - Session 6. Under the stated output-by-input convention, compute the Jacobian of $g(x,y)=(x+y,xy,x^2)$ and give its shape.
- 7 N18 - Session 7. For the objects below, compute $(P+P^T)x$.

$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad P = \begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix}$$

- 1 N19 - Session 7. For $u=x^2$, $v=u+1$, $L=v^3$, run forward and backward passes at $x=2$.
- 2 N20 - Session 8. Find and classify stationary points of $f(x,y)=x^2-y^2$.
- 3 N21 - Session 8. Find the second-order Taylor polynomial of $f(x,y)=e^x \cos(y)$ at $(0,0)$.
- 4 N22 - Session 8. Minimize $f(x,y)=2x^2+y^2-8x+6y$ and determine whether the conclusion is local or global.
- 5 N23 - Integrated. A 3 by 2 data matrix has singular values $5,1$. Find its rank and best rank-1 errors in both norms.
- 6 N24 - Integrated. For $f(x)=x^4$, compute first and second derivatives at zero and explain what extra analysis is required.

Historical-Source-Aware BITS Questions

No item below is claimed as a verified past-paper question. The accessible preview is hosted by a third party and its date conflicts with another preview, so the potentially useful concept has been independently rewritten.

H01 - Distinct eigenvalues and invertibility

Classification: Authored BITS-style | Institution: Birla Institute of Technology & Science, Pilani - WILP | Code: target AMLCS ZC416; preview shows related AIMLC ZC416 spelling | Term/year: authored for 2026; inspiration preview says Sem 1 and 18 Jan 2025, uncertain | Hosting: third-party, not an official archive | Source: [Scribd preview](#)

A real square matrix has distinct real eigenvalues. Decide which conclusions must follow: diagonalizability, invertibility, both, or neither. State the additional condition needed for any conclusion that does not always follow, and give a 2 by 2 counterexample.

H02 - Rank and solution structure

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For an m by n system $Ax=b$, write a decision rule using $\text{rank}(A)$, $\text{rank}([A|b])$, and n that separates no solution, one solution, and infinitely many solutions.

H03 - Hessian decision limits

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At a stationary point of a twice-differentiable real function, compare the conclusions available from positive-definite, negative-definite, indefinite, and semidefinite Hessians. Identify which are local conclusions.